Machine Learning in Finance

k Project # 3

Scenario

The team did so well on the previous two projects that the portfolio strategists want more! Your group has already presented a half-dozen ideas. The strategists came back with three big questions:

1. How do we know the models have the right parameters?
2. How well can the models be expected to work for predicting future cases?
3. How can the models be used together?

Responding to these tasks will surely give the strategists confidence in your ability to use these models profitably. Let's consider each of these questions as issues.

The first question asks, "How do we know the models have the right parameters?" This question requires that we know how to optimize hyperparameters. This means we'll have to explain which hyperparameters are used, how we optimize them, and how we know they are optimal.

The second question asks, "How well can the models be expected to work for predicting future cases?" This question requires that our models work well both within-sample—on training data—as well as out-of-sample—on testing data. Bias measures the accuracy on training data, and variance measures the accuracy on testing data. As such, this question is motivating us to discuss the process of optimizing the bias-variance tradeoff.

The third question asks, "How can the models be used together?" This question requires us to think about the various ways models can be combined. Some methods exist that combine the same type of model—homogeneous ensemble methods. Other methods exist that allow us to combine different types of models—heterogeneous ensemble methods. We want to show that, when combined, multiple models can sometimes outperform individual models. This will motivate us to show how models can be used together.

So, in summary, the three topics to be addressed are:

- Issue 1: Optimizing Hyperparameters
- Issue 2: Optimizing the Bias-Variance Tradeoff
- Issue 3: Applying Ensemble Learning—Bagging, Boosting, or Stacking

However, each topic will have two sections: a technical and a non-technical section. Be careful not to mix the two up. For example, if you decide to include equations and code, those should go into the technical section. Points will be deducted if those are in the non-technical section.

Within each section, you will select the relevant areas. You may use definitions, equations, etc. Be sure that the non-technical section is very specific. It should answer each of the three questions posed above.

Tasks

Step 1

members read through the feedback from GWP2. All group members will incorporate the suggested corrections on the three items and <u>work together to improve the questions submitted from GWP2.</u>

Step 2

all three members plan a strategy as to what goes in the technical and non-technical section of each question. The deliverable from this section will be an outline.

- Issue 1: Optimizing Hyperparameters
  - Technical
  - Non-technical
- Issue 2: Optimizing the Bias-Variance Tradeoff
  - Technical
  - Non-technical
- Issue 3: Applying Ensemble Learning—Bagging, Boosting, or Stacking
  - Technical
  - Non-technical

Step 3

each student writes 3–5 pages addressing their issue. Be sure that the title of each section (technical, non-technical) is in a large font and in bold face. Within each section, use bullets (e.g., description) that are from the outline in the previous step.

Step 4

each student reviews another student's part. Here, be sure to check accuracy, clarity, spelling, etc.

	Wrote	**Reviewed**
Student A	Issue 1	Issue 3
Student B	Issue 2	Issue 1
Student C	Issue 3	Issue 2

Step 5

Revisit GWP1 and update the marketing material that persuades potential investors that the ML tools will add value.